



PROJECT MUSE®

Considering Health Care Needs in a Rural Parkinson Disease Community

Reshmi L. Singh, Erin J. Bush, Mary Jo Cooley Hidecker, Catherine Phillips Carrico, Sandra Sundin



Progress in Community Health Partnerships: Research, Education, and Action, Volume 14, Issue 1, Spring 2020, pp. 15-28 (Article)

Published by Johns Hopkins University Press

DOI: <https://doi.org/10.1353/cpr.2020.0005>

➔ *For additional information about this article*

<https://muse.jhu.edu/article/753433>

Considering Health Care Needs in a Rural Parkinson Disease Community

Reshmi L. Singh, PhD, MS, BPharm¹, Erin J. Bush, PhD, CCC-SLP², Mary Jo Cooley Hidecker, PhD, MS, MA, CCC-A/SLP², Catherine Phillips Carrico, PhD³, and Sandra Sundin, MBA⁴

(1) University of Wyoming—School of Pharmacy; (2) University of Wyoming—Division of Communication Disorders; (3) University of Wyoming—Wyoming Center on Aging; (4) Wyoming Parkinson Project Community

Submitted 27 February 2019, revised 19 September 2019, accepted 27 September 2019

Abstract

Background: People living with Parkinson disease (PD) have multiple health care needs that intensify over time, because the disease is both chronic and degenerative. Past research indicates that issues with mobility, financial constraints, and lack of support networks impede access to health care for people with PD. These challenges are elevated for individuals who live in rural communities due to the lack of local health care professionals and specialists and support resources, and the need to travel to see providers/specialists. The research objective was to have PD community stakeholders identify health care barriers and resources as well as possibilities for improved health care in a rural state.

Methods: Focus groups were conducted in the context of a community-based participatory research (CBPR) approach. Focus group data collection helped create comfort and parity in the discussion, while a CBPR approach allows for authenticity of the findings because members of the community in question are involved as researchers. The responses were recorded and transcribed verbatim. Coding and organizing

of themes was completed manually and using NVIVO 10 software.

Results: Qualitative analysis revealed three main themes, PD issues, access issues, and stigma. These themes described disease-related stigma and concerns about disease progression and treatment, as well as challenges in accessing information, providers, and support by the PD stakeholders. The study results provided insight into the needs of people living with PD in rural communities.

Conclusions: Rural PD stakeholders proposed the use of technology (e.g., telehealth) to provide support to improve health care for people with PD.

Keywords

Allied Health Occupations, Community-Based Participatory Research, Rural Health Services, Rural Population, Parkinson's Disease, Qualitative Analysis

PD is the second most common neurodegenerative disorder and affects at least 1 million people in the United States.¹ With no known cause or cure, persons with PD (PwP) use prescription medications and behavioral interventions to alleviate key problems such as difficulties walking, handling objects, and speaking. In general, PwP may have only 39% adherence to medication use suggesting a need for medication management from pharmacists.² Difficulty with movement, particularly walking, is a hallmark symptom with PwP that need skilled

physical therapy.^{3,4} Seventy to ninety percent of PwP develop speech problems, but fewer than 20% report receiving speech treatment.⁵⁻⁷

PwP in rural areas may find these difficulties exacerbated. Although no U.S. research has been published on the experience of living rurally with PD, some rural issues have been suggested in research from other countries, including Australia,⁸ Bulgaria,⁹ and Croatia.¹⁰ For example, rural living, as compared to metropolitan residence, was a significant negative predictor of health-related quality of life in PwP in

Australia.¹¹ Also, since few PD specialists practice in rural areas PwP must choose whether to use local generalists to manage PD symptoms, travel to see specialists, or forgo PD specialized health care. Such issues may also increase rural PwPs' difficulties with managing this complicated, chronic disease.

Health care needs of PwP in Wyoming had not been investigated before this study, to the best of our knowledge. Wyoming is a rural, frontier state with low population and great distances between towns.¹² Thus, the researchers sought to understand the unique community perspective of these rural PD stakeholders. CBPR is a systematic approach to bring together a community of stakeholders to identify and improve health care issues.^{13,14}

The specific research objective of this study was to have PD community stakeholders identify health care barriers and resources as well as possibilities for improved health care in a rural state.

METHODS

Design

We conducted this study using a qualitative description approach. The philosophical underpinnings of our study were consistent with the six principles outlined by Bradshaw, Atkinson, and Doody¹⁵ in their important work identifying methodological aspects that guide and enhance the rigor of qualitative description. Those philosophical tenets are represented in our study and underscored in what follows: a) our process was exploratory and inductive, in that we gathered information and then allowed concepts and themes to emerge; b) we recognized the subjectivity of participants, as well as the researchers, through a social constructivist lens, acknowledging that there is not one reality and that individuals, through interpretation of experiences, create meaning; c) we sought to explore and understand stakeholders' experiences, rather than relate them to an extant theory; d) the researchers took an active and integral role in the study and acknowledged their influence on the study, fitting particularly well with a CBPR approach; e) interpretation, to some extent, is needed to identify emergent patterns in the data, make connections between them, and then disseminate how participants explained or ascribed meaning to their experiences¹⁶; and f) our study

was paradigmatically influenced by naturalistic inquiry, as we sought to study the perspectives of PD stakeholders in their natural form.¹⁷ We also want to acknowledge here the application to the community health development principle of community participation and control that describes the client group having expertise on their community culture and hence they are equal partners with health groups in community development planning and solutions.¹⁸ Qualitative description does not have specified sampling procedures, stipulated data collection, or specific analysis procedures.¹⁹ Rather, it is a general method used to derive meaningful results. Studies employing qualitative description require less interpretation during analysis and explanation of findings as results are typically "data-near."²⁰ Therefore, this qualitative approach is used when the results obtained are expectedly closer to the data and in need of less transformation than other methods.²⁰ Data-near results makes the process of reaching consensus more efficient than data requiring interpretation.¹⁹ Our study aimed to derive a deep understanding beginning with literal interpretation.²¹ The benefit of descriptive results for our study is reflected by our aim—to represent findings in a transparent way to expand existing resources in this rural state and to inform future efforts of increasing and optimizing available resources. Sandelowski^{20,21} describes qualitative description as including a wide range of processes and techniques, yet despite these undetermined procedures it should also incorporate techniques that are sensible, chosen judiciously, and well-informed by qualitative research principles.

The focus group methods and findings described herein are part of a larger study, of which a portion of the results has been previously published.²² The original study purpose, described herein, was to query stakeholders about the needs and resources, regarding PD, in their rural state. Our previously published articles^{22,23} describe the outcomes²² and methodology used²³ for a separate analysis solely addressing the resource of support groups. A separate analysis was conducted for support group information because of the magnitude of data provided on the topic as well as the categories and concepts that emerged from our preliminary analysis. The current article provides the opportunity to highlight the CBPR approach and provide descriptive data relaying the barriers and resources communicated by the PD stakeholder focus groups.

Researcher Groups

For this study, we had four groups of researchers. Our four groups of researchers were community researchers ($n = 4$), university researchers ($n = 5$), a representative of a community organization ($n = 1$), and student researchers ($n = 8$). Figure 1 displays the CBPR roles for each group and the collaborative outcomes at each phase of the study.

The group we refer to as community researchers included a person with PD and his caregiver (author SS), who initially were awarded the first phase of a pipeline grant from the Patient Centered Outcomes Research Institute. Community researchers had the choice to participate in or moderate focus group discussions. All community researchers that had PD opted to be participants in the focus groups, which fits a CBPR approach, as this approach calls for individuals who are in the community or group of inquiry to participate in the study in multiple ways.

Thus, the project started with community researchers and they are seen as the founding group of researchers (Figure 1). After being awarded the initial community engagement grant, they began forming a research group by involving a retired university professor/caregiver and another PwP. These

individuals formed the Wyoming Parkinson Project (WPP) team. After being awarded the first grant, the WPP aimed to find university researchers that could serve as coordinators of the project, mentors to the community researchers, as well as manage and analyze data and disseminate findings from the project. Interdisciplinary university researchers, which included authors RS, EB, and MJCH and later a project coordinator, responded to the solicitation and joined the community researchers to form a governing body of the WPP. Soon after, a community organization, the Wyoming Center on Aging expressed interest as well and their associate director (author CC) joined the project to represent Wyoming Center on Aging. University researchers offered undergraduate and graduate research laboratory students an opportunity to participate in the project, and several students volunteered. The following two phases of the Patient Centered Outcomes Research Institute pipeline funding were written by and awarded to the entire research group.

The WPP was formed to engage and connect PD stakeholders with the research community to assist in identifying and working to improve PD health outcomes in this rural state. Researchers and community partners recruited PD

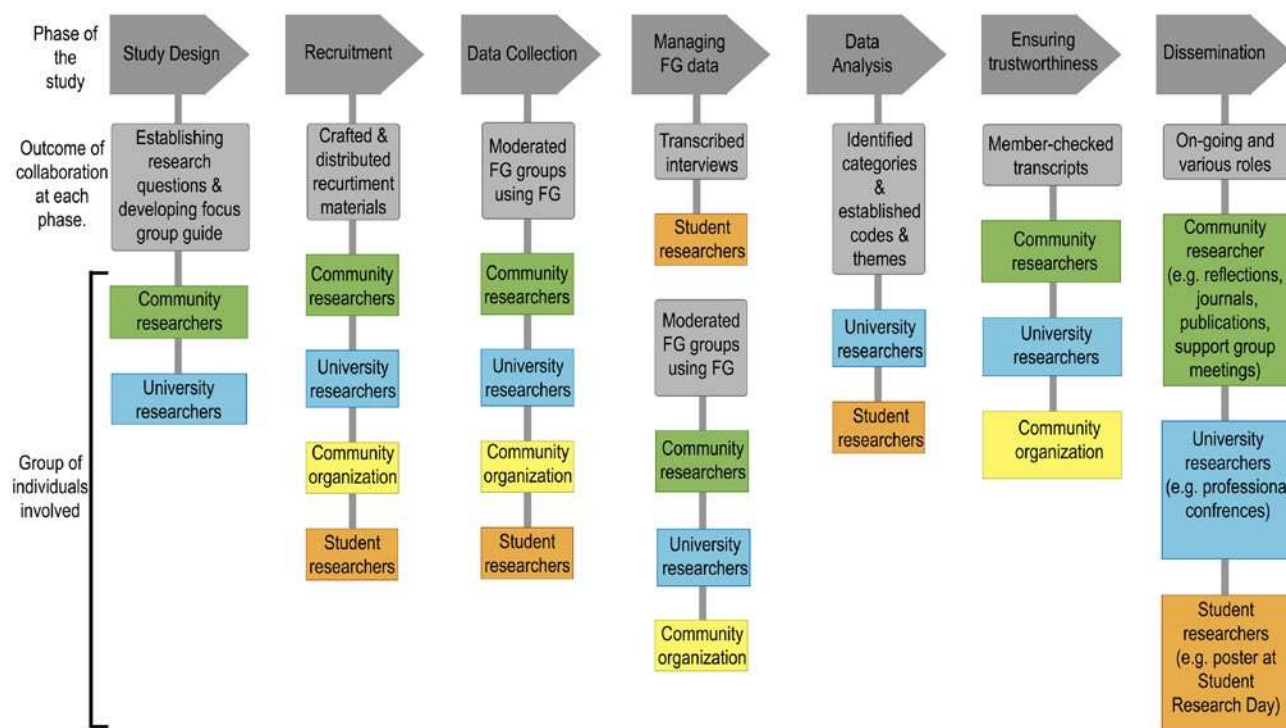


Figure 1. CBPR roles and outcomes for each phase of the Parkinson disease study

stakeholders to participate in the study herein, by sending out a call for participation through the WPP newsletter, and then by word of mouth and snowball recruiting. The call for participation invited PD stakeholders to attend a free luncheon meeting (funded by the Patient Centered Outcomes Research Institute grant), to inform Parkinson's stakeholders about the WPP's purpose and goals. After the luncheon, all stakeholders were invited to stay for focus groups so that they may give input about needs and resources for PwP in Wyoming. The luncheon was held in a centrally located town in the state, and participants (who were from different parts of the state) were informed, before the luncheon, that they could attend via video conferencing if they preferred or if they were unable to attend in person.

Participants

Thirty stakeholders, PwP, caregivers, audiologists, occupational therapists, pharmacists, physical therapists, social workers, speech-language pathologists, and other public health professionals from throughout the state, participated in five focus groups. Each focus group included four to nine individuals; one focus group participated by video chat (four individuals). A moderator for each group presented general open-ended questions and asked follow-up, context-dependent questions. The focus groups demonstrated minicommunities, as the researchers sought to represent community interactions and perspectives of diverse stakeholder backgrounds. Each focus group consisted of PwP, caregivers of PwP, as well as health care providers who had a focused interest in treating PwP.

We asked individuals to sit where they were most comfortable for the focus group discussions, and provided an opportunity to socialize before the focus groups, helping to balance the power relationships between researchers and participants.^{15,24} We conceptualized our focus groups as minicommunities because participants chose the group he or she participated in for lunch and subsequent focus group discussions. Sitting where they were most comfortable may have led to more open sharing about topics than would have been achieved by focus group assignments. Also, participants were all members of the larger community of PD stakeholders in our rural state; therefore, these small groups mirror the actual composition of the local PD community.

Post hoc, we were able to further assess that these groups represented mini-communities because there was a relatively equal distribution of caregivers, people with PD, and professionals, in each group that occurred naturally. Additionally, the information shared about their opinions and perspectives were very similar in content both within and across groups. Finally, although this was not instructed or implied, the PwP seemed to be revered as the experts in the individual focus groups. Questions were often diverted by other members of the focus groups to them or, in instances where their communication was too impaired, to their caregiver. In the one participant-caregiver dyad that this occurred, the caregiver seemed to help the PwP communicate his or her thoughts and feelings, and used a supplemental communication board to allow the participant to indicate letters or speaking of the intended word, providing an alternate way to express his or her message.

Procedures

The community researcher group, during initial planning meetings, expressed a desire to hear from as many PD stakeholders as possible. Thus, data collection using focus groups was chosen. Focus groups maximize the number of participants in the study and are often thought of as less threatening than one-on-one interviews for people to share thoughts, ideas, and opinions.²⁵ Typically, three to six groups are viewed as sufficient to obtain theoretical saturation in a single category.^{24,25}

Participants attended focus groups that were facilitated by moderators and co-moderators to discuss the quality of care for PwP in Wyoming. Moderators were some of the community researchers and the university researchers. Comoderators were some of the community researchers and the student researchers (Figure 1). Each moderator used a focus group discussion guide that included open-ended questions and following that stimulated discussion about issues previously identified by the community researchers. The open-ended questions centered on the barriers to health care as well as the existing resources. This document (Appendix) was used as a guide for discussion as opposed to a focus group protocol. The guide was developed using agreed upon issues expressed by the community researcher group. The community and university researchers collaborated on the

development of the guide, ensuring that the areas the community researcher deemed important, received focus. We sought specific information that was pertinent to this rural population, to affect change for them, as opposed to gaining theoretical knowledge of general barriers and resources affecting PD communities. Because our approach was CBPR and a future aim is to incite resources and ameliorate barriers for these individuals, we did not use studies about other rural communities to inform the development of our guide. After the three open-ended questions were discussed during the focus groups, we asked the participants to rank order the importance of the issues (list developed previously with coauthor and community partner SS) according to their experiences. Rank ordering is used as a strategy to facilitate further discussion in mixed groups.^{24,26} Focus group discussions were audiotaped and transcribed verbatim by student researchers. Each moderator checked the final transcripts (Figure 1). Deidentified data was analyzed manually and organized in NVIVO10 (QSR International, Doncaster, Australia) qualitative research software.

Data Analysis

A general data analysis process fit well with our non-categorical approach of qualitative description, as well as the exploratory nature of the study's subject matter. As Sandelowski^{20,21} suggested, we used multiple qualitative strategies that fit our study's aim and collected data. Throughout the data analysis, various qualitative procedures were used. That is, we used open and axial coding procedures. Additionally, because our procedural focus was informed by CBPR principles, we queried multiple types of stakeholders.

Carter et al.²⁷ describe data source triangulation as "the collection of data from different types of people including individuals, groups, families, and communities." Our data source triangulation centered on using multiple sources (number of participants as well as types of participants) to obtain a wide breadth of experiences and achieve data saturation.²⁷ Investigator triangulation was also accomplished through collaborative discussion of our observations from conducting the focus groups as well as our consensus regarding themes with all study authors.^{26,28}

Data saturation was reached as we solicited participation from every PD stakeholder in the state. Additionally,

we believe saturation was achieved by coding until there was no longer practicality in coding further, and our interview questions were designed to query multiple participants, and types of participants (i.e., caregivers, people with PD, professionals who work with individuals with PD).²⁸ Fusch and Ness²⁹ explain that conducting focus groups is a way to elicit multiple points of view from a large pool of potential participants, thereby reaching data saturation.²⁹

For the analysis of unabridged transcripts, three authors (RS, EB, and CC), who have qualitative research expertise first conducted topic coding, followed by analytic coding.³⁰ Topic coding was used to identify the main concepts that emerged from the data. Analytic coding followed, which expanded our identified topic codes. This was done through interpreting the data and reflecting on the meaning of the data by authors RS and EB. During analysis, other groups of researchers were informed and received updates about the analysis.

Upon completion of the data analysis, a summary of the results was presented to the rest of the WPP governance body. Checking and reviewing study results by all coauthors helped to ensure study trustworthiness and rigor. Results were then shared with PD stakeholders during follow-up stakeholder luncheons held in two locations in the state. This opportunity provided member checking of themes and the vetting of ideas with PD stakeholders from across the state regarding future research. No significant changes to our findings occurred as a result of the member checking; however, some themes and terms such as patients were reworded to PwP. Dissemination of results from this project have been presented by various members of the researcher teams (Figure 1), published in WPP newsletter, and are ongoing.

Ethics Approval

Human subjects' approval was obtained through the university's institutional review board.

RESULTS

The demographic information for the participants is presented in Table 1. The stakeholders were PwP, caregivers and health care professionals, along with researchers and students involved in health care. Participants were all non-Hispanic, 50% females, 83% college-educated, and 37% were over 65 years old.

Table 1. Demographics of CBPR Focus Group Participants

Characteristic	N	Percentage
Sex		
Female	15	50.0
Male	14	46.7
Not Reported	1	3.3
Age		
18–44 years	9	30.0
45–64 years	8	26.7
65 + years	11	36.7
Not Reported	2	6.7
Education		
High School Diploma or General Education Development	5	16.7
College Degree	25	83.3

The main themes were around challenges dealing with PD in a rural state, existing sources of support, and possibilities or solutions for some of the identified challenges. Following analytical coding, we named these three main themes that were impacting rural PD health care as: (1) PD issues, (2) access issues, and (3) stigma.

The first theme, PD issues, described concerns related to the disease process and its treatment. Specific concerns were related to medication use, unique features of PD, and exercise. The second theme, access issues, generally described challenges in accessing information, providers, and support. Specifically, many stakeholders described a lack of coordinated care, lack of access to providers, travel necessity, limited access to support groups, and financial support challenges. The final theme was about the experience of stigma with PD diagnosis and treatment. Next, the three themes are described in detail along with text exemplars.

PD Issues

This theme was further dichotomized into the challenges of chronic illness and the challenges unique to PD. The challenges of chronic illness were those that would be commonly experienced by many individuals with a chronic and/or progressive disease, like PD. These topics included medication and exercise challenges.

Medication-related issues emerged as a challenge from the stakeholders. These primarily included not having enough information about the medications and not being able to easily

get counseling regarding changes in medication. The idea of a “medication helpline” was presented by some as a solution to address these issues while reducing the need for travel. The following excerpt came after describing a serious issue of not getting support or counseling from a pharmacist regarding medications for PD.

It would be good to have a [pharmacist available] . . . Maybe a special pharmacist because the balance and the dosage is just really mind boggling. You go higher [in dosage] and then . . . you get fluid in your lungs and in your feet . . . so it's just hard to determine the correct dose.

This PwP also shared that he had cut a medication patch in half to try to reduce the dose. Participants also discussed challenges they believed to be unique to PD, that is, PD-specific features. One focus group member mentioned that there were at least “27 combinations of symptoms” or ways that the disease manifests. Most caregivers and PwP agreed with the comment. An additional point was that not everyone shows the same signs and symptoms of PD (e.g., tremors, balance problems). This diversity of symptoms can lead to a lack of or a delay in PD diagnosis was brought up by many respondents. This idiosyncratic nature of PD was further illustrated in this statement:

Yeah every person [with Parkinson] is different. Like myself, I fall backwards real easy. . . . one of the ladies down there said her stomach gives her real fits . . . every one of us [has] different symptoms . . . If you've met one [PwP] you've [only] met one [PwP].

Different ideas about what caused PD were discussed as a separate topic that the researchers had not specifically queried from the focus group. Caregivers and family members especially wanted to know what had most likely caused this disease in their loved ones. The following quotations addressed a common discussion regarding PD potential causes. “And another friend . . . he's got Parkinson's. We all worked in the same refinery. There might be other guys who worked there with us but we don't know.” “It might be related to the refinery.” “A couple of those guys that come down to our mines they worked in radiation.” “Yeah, I think there's a lot more people in Wyoming [with Parkinson's] . . . if there's a

connection with radiation, if there's a connection with oil refineries, there's a lot more people out there [with PD] than we even know about."

Other than working in a refinery, having been exposed to Agent Orange during the Vietnam War was brought up as a potential risk factor by participants.

Exercise-related issues for participants were mainly regarding motivation or ability to exercise depending on severity of PD symptoms. Motivation was addressed by many by who suggested creating exercise groups, either stand-alone or as part of established support groups in their community. As one PwP mentioned, "I think people with Parkinson would go if there was a group," suggesting it would make exercise more fun. Some stakeholders mentioned the possibility of online exercise groups and while this brought up further questions regarding "cost and ease of online groups" it was also a way to "keep it interesting" for PwP. Most respondents agreed that it would be a great idea for those who are comfortable with technology and have an Internet connection.

Access-Related Issues

This theme described issues related to a) access to information for PwP and caregivers, b) a lack of coordinated care from primary and allied health care professionals, and c) a lack of community support and financial support. Access to information for individuals included mention about the need for education regarding PD, as well as PD care and management. "I think just access to information [is a need]. As a PwP, I have had really hard time finding information about what's happening and what I can do about that." As demonstrated by the following example, educating PwP and caregivers is needed, perhaps through the use of technology via Internet access and websites. Increasing support group access may reduce PwP isolation.

"There's just a challenge of getting information out . . . because once you get diagnosed with [PD], then where do you go for the info? . . . [the WPP] started a website with info . . . You still have to get that information out to providers or to support groups . . . how do you hook up with people who are so isolated?"

One of the PD stakeholders brought up the need for training allied health care providers so they may offer expert advice

on PD related care. This individual, referring to a specific therapeutic program, called Lee Silverman Voice Therapy (LSVT BIG), felt that specific treatment programs, such as this one, should have more exposure, garnering more trainees with expertise in PD.

once we start seeing patients with PD, that's why I sought out LSVT [BIG training] for Physical Therapy . . . So I think actually allied health, especially in rural communities can be a great resource for people . . . , we're at least provided an opportunity to make ourselves a little bit more of an expert to then help facilitate the patient's treatment.

Additional comments were made by allied health care providers, particularly regarding the greater accessibility of allied health care versus primary care.

Access issues included lack of coordinated care which was described as a lack of providers/specialists and the need for allied health care providers to manage PD care coordination in a rural community. This was an important point of discussion by each group since they felt the PD related cognitive issues demanded additional help from a variety of care providers.

I think that . . . there's so much coming from every direction. . . . and I think they need someone to help coordinate all of this. To put it all together because it's so much for them—especially if they're going through some cognitive declines as well. It's really hard to manage all this information."

Discussion of technology as a means for potentially coordinating and improving access to care was brought up by some participants wherein the LSVT was suggested as a telehealth option to provide both speech therapy and physical therapy for PwP.

We need more . . . physical therapists and occupational therapists who are familiar with Parkinson's specifically. And particularly this program which [we] are really sold on, if there were more people who were certified in the LSVT physical therapy and speech therapy.

The need to travel long distances for health care was discussed frequently. In addition, participants mentioned that being

in a rural state, combined with mobility issues of PD made it difficult for caregivers and PwP to access care. “Just the distances [are] hard. I mean, it takes a lot of energy for all of us . . . if you don’t live close [to providers]. It takes all day to do physical therapy, or you know, just life. It’s exhausting. And Wyoming makes it harder.”

Support groups for PwP and caregivers were noted as a resource; however, some PwP had limited access. The study participants appreciated the availability of support groups to receive information and connect with their PD community. The limited nature of this type of support was due to the long distances needed to travel to groups if there was not one nearby. Information about support groups is presented here as it was identified as a resource as well as an example of access issues. Because of the volume of information present in our data corpus regarding support groups, this content is represented as a separate study, undergoing more in-depth analyses, and distinctive results.²² The methodology of this analysis is also published elsewhere.²³ Aspects of support group findings, however, are briefly described herein because they are essential in gaining a holistic view of barriers and resources described by stakeholders. One way to improve this access issue was “online support groups,” suggested by some stakeholders.²²

Financial barriers such as the high cost of treatment and lack of insurance coverage led to study participants reportedly not being able to initiate or continue care on a regular basis. Similarly, from the health provider perspective, it was harder to help PwP who did not have health insurance since the cost of PD treatment can be prohibitive.

One of the people I talked to on the phone that has Parkinson, has had it over ten years . . . but she hasn’t seen a doctor in ten years, because she . . . doesn’t have the money, nor the people to help her get to see a doctor, so that tells you what kind of things we’re working against.

The following example of comorbidity of depression and the issue of receiving inconsistent therapy was brought up by a PwP. Physical symptoms of PD, apathy and depression, coupled with the lack of access to a therapist, significantly multiplied the burden of PD. This was an important realization that brought together the PD issues and access issues.

I also have a therapist who works with me . . . helps me adjust to what’s happening to me. And my spouse also sometimes comes with me. . . . She’s kind of had to learn too. She thought I was getting really, really depressed . . . [because] when you get Parkinson you get tired sometimes, you can’t talk very well, you’re soft and you don’t seem to have any strength. She says, “I’ve never seen you so down.” I have a tendency to be down. And, that’s another problem, for Parkinson people to find . . . therapists . . . who have worked with Parkinson before so they have some idea of what you’re going through and when you need different kinds of therapy.

The use of technology emerged as a solution to address the issues identified from data gathered from this rural PD community. In fact, regarding both the disease process and limited access, technology emerged from the community as a possible solution to overcoming issues. Technology may be used to increase access to a) physicians and allied health providers (e.g., telehealth) and b) education and information (e.g., webinars), thereby, potentially improving aspects of the disease process through better symptom management, more education, and greater access to support services.

Stigma

The final theme pertains to the issue of stigma. This theme, although smaller in terms of amount of pertaining data, was deemed worthy of its own theme given its significance. As one participant noted about undiagnosed PwP, “They just don’t want to go to the doctors. They’re in denial.” Identified reasons for stigma were multifaceted in PwP—sometimes stigma resulted from diagnosis in younger persons while other times the stigma was fear of death or needing assisted living accommodations.

The following example relates to stigma resulting from lack of awareness of the serious nature of PD or as is commonly seen—a person’s denial that their condition necessitates assessment/treatment by a health care professional.

Well it could be in a small community like [name of place], just sheer embarrassment. We older generations especially, anything that’s a little bit out of the ordinary, you don’t necessarily want to admit it. I

know a gentleman . . . who has Parkinson's. I mean, you can tell that he does, but he doesn't . . . want to admit that he has Parkinson's or be different than anybody else. Maybe education will help that.

Furthermore, the stigma may result in denying/refusing help. This stakeholder had previously indicated a desire to share information with others about PD. "Well, I'm in [name of place]. There are people with Parkinson's there, but my experience has been that they are very reluctant to . . . discuss anything."

DISCUSSION

Themes revealed through the current study indicate that rural residents diagnosed with or caring for a PwP experience confusion, uncertainty, and isolation. Additional education about the disease, its stigma, and access to available treatments and support resources, as well as consistent access to coordinated health care is a significant need of PwP in this rural area. The use of technology to facilitate communication with health care professionals, access to therapy and exercise programs, learning about the disease, and participation in support and networking groups were all solutions suggested by participants to overcome the barriers inherent to living in a rural location.

Symptoms caused by PD often worsen as the person ages.¹ With no known cause or cure, PwP's disease use prescription medications and behavioral interventions to alleviate key problems such as walking, handling objects, and speaking. Without ongoing coordinated services, PwP may become isolated due to absent or ineffective medication and behavioral interventions, leading to a decreasing quality of life.^{8,31} PwP, who accessed a multidisciplinary allied health care intervention, showed functional gains.³¹ This same need for multidisciplinary care was established in a previous study examining palliative stage for PwP and family experiences.³² Furthermore, the prevalence of PD is purported to nearly double in the next 25 years.³³ As the number of PwP increases, care must be efficient and well-coordinated.

The Centers for Medicare and Medicaid Services has established billing codes intended to reimburse health care practices for care coordination of individuals with chronic conditions, which can include PD.³⁵ Chronic care management can be

conducted by a primary care provider office or a specialist such as a neurology practice, which could be of particular benefit to people living with PD.³⁵ Chronic care management could serve as a useful facilitator to improve the fragmented care that participants in the study report experiencing.

The use of technology to deliver medical and allied health services to those living with Parkinson is a growing area of investigation that is particularly relevant to the results of this study. A review of published prospective design telemedicine studies for treatment of PD indicates that telemedicine is a feasible modality for service delivery, although the case for efficacy of telemedicine for treatment of PD or increased access to allied health care has not been made yet.³⁶ Echoed in the results of the current study are the barriers associated with use of technology to facilitate telemedicine (i.e., access to broadband Internet and lack of familiarity with computers or smart phones). As the population ages, older adults who develop PD will likely be increasingly familiar with the Internet and technology, which should alleviate some of the current barriers experienced by older adults.³⁶

Allied health services have been shown to be effectively delivered via telehealth to rural and remote residents.³⁷ The LSVT-LOUD speech therapy program shows feasibility when piloted via telehealth delivery to PwP.³⁸ Access to allied health services can be extremely limited in rural and frontier areas. The ability to offer evidence-based interventions remotely is a powerful tool to extend the reach of crucial interventions shown to improve the functioning of PwP. The participants in the current study indicate that delivery of medical and allied health services would be acceptable to rural residents and could ease the burden of living with Parkinson in a rural area while improving the health care of the PwP.

Another promising application of technology is increasing access to education and support, which is a significant need identified in the current study. As our participants noted, the Internet could serve as a gathering point for support and/or education groups. Online health communities use social media platforms to bring together people with a shared interest in a particular health condition including people living with the condition, caregivers, health care professionals and researchers.³⁹ The primary functions of online health communities are to 1) facilitate the exchange of medical experience and knowledge, 2) enhance interdisciplinary collaboration

across institutions, 3) provide a platform to support self-management, and 4) improve patient-centered care.³⁹

The use of Online Health Communities may be of particular benefit to PwP in rural areas, as there are limited opportunities for education, fewer opportunities for interdisciplinary care, and less access to chronic disease self-management education.

FUTURE DIRECTIONS

Results of this study inquiring into needs of rural PD stakeholders directs our future plan to continue using CBPR, to build on capacity and engagement, and identify potential comparative research questions for PwP in our rural community. Our forthcoming research trajectory has the ultimate aim of continuing involvement of the WPP members to plan useful and accessible interventions such as the use of telehealth to deliver allied health care, ultimately improving quality of life for the PD community in our rural state and beyond. Last, it is noteworthy that the state university was pointed out as a resource accessible to the entire state, and a possible means to make contact with experts including pharmacists, speech-language pathologists, and exercise specialists as well as technology resources.

LIMITATIONS AND CHALLENGES

Five focus groups were conducted and hence results are not generalizable to individuals who did not attend or to other rural PD communities. This form of qualitative data collection is most useful for a smaller sample size. Our results partially mirrors those from the rural health care delivery study for PwP in Australia.⁸ This indicates that some needs for PD care, rurally, might be similar across the world. To facilitate discussion in the focus groups, participants were asked to rank order the importance of PD-related issues according to his or her experience. It is possible that the approach of a predefined list of potential PD-related issues could hinder participants from bringing up other, undefined concerns. To minimize this effect, participants were offered the opportunity to identify other concerns during the open-ended discussion before the rank ordering was presented. A mixed group of stakeholders participated in each focus group, however, no physicians, neurologists, or state/federal public health officials

participated. The WPP team plans to engage physicians and public health officials in the state in the future. Because this was a CBPR study that used focus groups, extended time was needed to analyze data and conduct member checking.

CONCLUSION

This preliminary, qualitative, community-based study brought together several PD stakeholders to discuss issues most pertinent to pursue with further funding and research. The present study expands information on rural PD community health care needs by using a CBPR approach to further understand not only needs, but also solutions, to improve care and hence quality of life from rural PD stakeholders' perspectives. Such a method allowed the community members to be 'change agents' by not only sharing their stories and perspectives about issues pertinent to PD, but also recommending strategies for addressing these issues.¹⁴ One common solution proposed was use of technology that would potentially address multiple issues, such as access (telehealth), education (webinars), and stigma (online support groups) for rural PD stakeholders. As far as we know, this was the first, in-depth community engagement effort in the United States for understanding care for patients with PD living rurally.

ACKNOWLEDGMENTS

The Wyoming Parkinson's Project (WPP) is sponsored by the Cheyenne PD Support Group-led by Christopher Herron, PhD, a person living with Parkinson's disease (PwP), and Sandra Sundin, MBA, his spouse and caregiver. Other members of the WPP that had a key role in the project described herein were Rex Gantenbein, PhD, a retired professor in telehealth and caregiver; Judith Powers, PhD, a person living with PD; Beth Young Jones, MS, project manager; and Ryan Rausch, OTR/L, an occupational therapist.

We thank the PD stakeholders who took part in the focus groups and shared their perspectives and experiences. Focus group moderators and co-moderators (in addition to the authors) were Rex Gantenbein, Christopher Herron, Beth Young Jones, and Judith Powers. Student researchers included Katelonne Adams, Clarissa Petres, Brittney Arevalo, Allison Long, Bree Olson, Adele Riley, Trudi Fisher and Noel Schatz.

REFERENCES

1. Parkinson's Foundation. Understanding Parkinson's: Parkinson's FAQ. Miami (FL): Parkinson's Foundation. Available from: <http://f5h3y5n7.stackpathcdn.com/sites/default/files/attachments/Parkinson%27s%20FAQ.pdf>
2. Davis KL, Edin HM, Allen JK. Prevalence and cost of medication nonadherence in Parkinson's disease: evidence from administrative claims data. *Mov Disord*. 2010;25(4):474–80.
3. Allen NE, Sherrington C, Paul SS, Canning CG. Balance and falls in Parkinson's disease: a meta-analysis of the effect of exercise and motor training. *Mov Disord*. 2011;26(9):1605–15.
4. Allen NE, Sherrington C, Suriyarachchi GD, Paul SS, Song J, Canning CG. Exercise and motor training in people with Parkinson's disease: a systematic review of participant characteristics, intervention delivery, retention rates, adherence, and adverse events in clinical trials. *Parkinsons Dis*. 2012;854328.
5. Sapir S, Ramig L, Fox C. Speech and swallowing disorders in Parkinson disease. *Curr Opin Otolaryngol Head Neck Surg*. 2008;16(3):205–10.
6. Simberg SR, Rae J, Kallvik E, Salo B, Martikainen K. Effects of speech therapy on voice and speech in Parkinson's after a 15-day rehabilitation course: a pilot study. *Int J Ther Rehabil*. 2012;19:273–86.
7. Atkinson-Clement C, Sadat J, Pinto S. Behavioral treatments for speech in Parkinson's disease: meta-analyses and review of the literature. *Neurodegener Dis Manag*. 2015;5(3):233–48.
8. Duncan GF, Rositano P. Parkinson's disease in regional Australia. *Rural Remote Health*. 2011;11(4):1658.
9. Hristova DR, Hristov JI, Mateva NG, Papathanasiou JV. Quality of life in patients with Parkinson's disease. *Folia Med (Plovdiv)*. 2009;51(4):58–64.
10. Klepac N, Pikija S, Kraljić T, Relja M, Trkulja V, Juren S, Pavlicek I, et al. Association of rural life setting and poorer quality of life in Parkinson's disease patients: a cross-sectional study in Croatia. *Eur J Neurol*. 2007;14(2):194–8.
11. Soh S, McGinley JL, Watts J, Iansek R, Morris ME. Rural living and health-related quality of life in Australians with Parkinson's disease. *Rural Remote Health*. 2012;12:2158.
12. U.S. Census Bureau. (2015). State Quickfacts: Wyoming. Available from: www.census.gov/quickfacts/table/PST045216/56.
13. Kieffer EC, Salabarría-Pena Y, Odoms-Young AM, Willis SK, Palmisano G, Guzman J. R. The application of focus group methodologies to CBPR. In: Israel BA, Eng E, Schultz AJ, Parker EA (eds). *Methods for community-based participatory research for health*. San Francisco (CA): Jossey-Bass; 2013;250–303.
14. De Marco M, Kearney W, Smith T, Jones C, Kearney-Powell A, Ammerman A. Growing partners: building a community-academic partnership to address health disparities in rural North Carolina. *Prog Community Health Partnersh*. 2014; 8(2):181–6.
15. Bradshaw C, Atkinson S, Doody O. Employing a qualitative description approach in health care research. *Glob Qual Nurs Res*. 2017;4: 2333393617742282.
16. Sandu A, Unguru E. Several conceptual clarifications on the distinction between constructivism and social constructivism. *Postmodern Openings*. 2017;8(2):51–61.
17. Lambert VA, Lambert CE. Qualitative descriptive research: an acceptable design. *Pac Rim Int J Nurs Res Thail*. 2012; 16(4): 255–6.
18. Steckler AB, Dawson L, Israel BA, Eng E. Community health development: an overview of the Works of Guy W. Steuart. *Health Educ Q*. 1993;20:S3–S20.
19. Cohen DJ, Crabtree BF. Evaluative criteria for qualitative research in health care: controversies and recommendations. *Ann Fam Med*. 2002;6(4):331–9.
20. Sandelowski M. Whatever happened to qualitative description? *Res Nurs Health*. 2000;23:334–40.
21. Sandelowski M. What's in a name? Qualitative description revisited. *Res Nurs Health*. 2010;33(1):77–84.
22. Bush EJ, Singh R, Hidecker MJC, Carrico CP. Parkinson's disease support groups in rural America: barriers, resources, and opportunities. *Qual Rep*. 2018;23(6):381–400.
23. Bush, E. J., Singh, R. L., & Sarah, K. (2019). Lived experiences of a community: Merging interpretive phenomenology and community-based participatory research. *International Journal of Qualitative Methods*, 18 doi:<http://dx.doi.org/10.1177/1609406919875891>
24. Krueger RA. *Focus groups: a practical guide for applied research*. Thousand Oaks (CA): Sage; 1994.
25. Krueger RA and Casey MA. *Focus groups: a practical guide for applied research*. London: Sage; 2009.
26. Morgan DL. *Focus groups as qualitative research*. Qualitative research methods series v 16. Thousand Oaks (CA): Sage; 1997.
27. Carter N, Bryant-Lukosius D, Di Censo A, Blythe J, Neville AJ. The use of triangulation in qualitative research. *Oncol Nurs Forum*. 2014; 41(5):545–7.
28. Guest G, Bunce A, Johnson L. How many interviews are enough? An experiment with data saturation and variability. *Field Methods*. 2006;18(1):59–82.
29. Fusch PI, Ness LR. Are we there yet? Data saturation in qualitative research. *Qual Rep*. 2015;20(9):1408–16.
30. Richards L. *Handling qualitative data: a practical guide*, 3rd ed. Thousand Oaks (CA): Sage.
31. Ellis T, Katz DI, White DK, DePiero TJ, Hohler AD, Saint-Hilaire M. Effectiveness of an inpatient multidisciplinary rehabilitation program for people with Parkinson disease. *Phys Ther*. 2008;812–9.
32. Giles S, Miyasaki J. Palliative stage Parkinson's disease: patient and family experiences of health-care services. *Palliat Med*. 2009;23(2):120–5.

33. Kowal SL, Dall TM, Chakrabarti R, Storm MV and Jain A. The current and projected economic burden of Parkinson's disease in the United States. *Mov Disord.* 2013;28(3):311–8.
34. Department of Health and Human Services. Connected care toolkit: chronic care management resources for health care professionals and communities. Available from: www.cms.gov/About-CMS/Agency-Information/OMH/Downloads/connected-hcptoolkit.pdf
35. Nuwer, MR. Chronic care management coding for neurologists. *Neurol Clin Pract.* 2015;5:430–4.
36. Schneider, RB, Biglan, KM The promise of telemedicine for chronic neurological disorders: the example of Parkinson's disease. *Lancet Neurol.* 2017;16:541–51.
37. Speyer R, Denman D, Wilkes-Gillan S, Chen YW, Bogaardt H, Kim JH, Heckathorn DE, Cordier R. Effects of telehealth by allied health professionals and nurses in rural and remote areas: a systematic review and meta-analysis. *J Rehabil Med.* 2018;50(3):225–235.
38. Donovan NJ. People with Parkinson's disease and dysarthria benefit equally from Lee Silverman Voice Treatment (LSVT LOUD) delivered via tele-rehabilitation compared with conventional face-to-face version of the treatment. *Evidence-Based Communication Assessment and Intervention.* 2017;11(3–4): 124–9.
39. van der Eijk M, Faber MJ, Aarts JW, Kremer JA, Munneke M, Bloem BR. Using online health communities to deliver patient-centered care to people with chronic conditions. *J Med Internet Res.* 2013;15(6):e115.

By Continuing to participate in this part of the meeting you are agreeing to be part of the focus group research which involves answering the questions in this guide and agreeing to be recorded for research purposes.

Freedom of Consent

Your decision to whether or not to participate will not affect your current or future relations with the University of Wyoming, your hospital or pharmacy, or your health care providers. If you decide to participate, you are free to not answer any question or withdraw at any time without affecting those relationships.

Questions about the Research

If you have any questions or concerns regarding this study or about your rights as a research subject and would like to talk to someone other than the researcher, please contact the University of Wyoming IRB Administrator at 307-766-5320.

If you do not consent, thank you for your time.

If you consent, please continue to answer I to IV below.

- I. Overall, what barriers exist in improving the quality of life of patients with Parkinson (PWP) in Wyoming?

- II. What are some resources/programs already in place in Wyoming for PWPs?

- III. Lastly, which key people, if any, are missing from this discussion?

- IV. Please rank order the following questions based on your perspective and experience:

- What care resources are available for PWP in Wyoming through direct and indirect methods such as pharmacy help lines, Talk-2-Talk, and telemedicine?
- What are the expectations of Parkinson's disease care for PWP and other stakeholders in the PD Community in Wyoming?

- What's more effective in providing support to rural PWP: face-to-face or online support? Either approach could include individual and group support.
- What topics or information can be effectively delivered in support groups? Possible topics might include education on PD symptoms, treatments, medications, assistive technology, possible disease trajectories; instruction on therapies for PD; and/or clinical information from practitioners.
- What are the most effective ways to get PWP to participate in support groups: access, topics, leadership, communication, practitioners, food, online/face-to-face contact, or other methods?
- Current medical and nonmedical resources available within a 50 mile sphere of each PWP.
- Current PWP knowledge of these available resources.
- Pharmaceutical burden each PWP experiences.
- Knowledge of the positive roles of diet and exercise in the overall health of the PWP, and their specific experiences in these two areas.
- State of knowledge about research within the national and international PD communities, and how this research may or may not impact their individual quality of life.
- PWP access to computer and/or the Internet. Would they be interested in novel uses of the Internet for things such as online PD support or exercise groups?
- PWP knowledge of, or any personal experience with, and access to, available adaptive/assistive technology.
- Has each PWP been given information about local or Internet based support groups? How did they access this information?
- What role do the specific PWP's financial, mental and/or physical situations play in accessing resources that might positively impact their quality of life?
- What would the specific role of a support group play in the individual PWP's quality of life? Has or could membership in a support group meet their needs? Why or why not?
- Use of providers as resources—physicians, pharmacists, specialists, etc.